

PLC NODE Real-protocol plant boundary				plc-node-simulator · film-line preset · 13/23 checks pass							
Modbus TCP live connect 30 ms		Modbus RTU live connect 16 ms		OPC UA live connect 17 ms		MQTT live connect 16 ms		HTTP live connect 17 ms			
Real sampling 9 points at ~895 ms cadence · 40 time-series points retained during the loop · link break sensed, reconnect ok											
GOVERNED Write path on the live link					SP → interlock → plant → readback · interdiction and freeze alarm are separate						
SP write 182 °C HTTP 200 in 25 ms		4-layer interlock hard range recipe window 5/5 attacks rejected		Plant model first-order lag PV-182 ≤ 3 °C reached in 30 s		Register readback v_rb - v ≤ τ τ = 0.7 °C accepted at 185 °C		F5 interdiction out-of-window writes blocked		F2 freeze alarm independent monitor alarm within 2 s	
readback closure does not cover process response											
AGENTTEAM Closed loop over the parameter layer					task board → time-range data → governed write → judged record						
1 Task board goal filed lead → worker		2 Time-range query daq_query from / to / bucket		3 Policy step scripted gains 0.5 / 0.35 / 0.3		4 Governed write dcw_control record opened		5 Judgment dcw_judge keep journal attributed		6 Task closed target verified COMPLETED	
RECORDED Two recorded missions					archive 20260920094610-to4 · phases P4m (twin) and P10 (line)						
Cast-film twin · P4m					BOPET line · P10					9 devices · 30 setpoints · 19 DAQ	
Objective melt temperature 200 → 204.704 °C					Thickness (μm)					26.90 → 26.82 → 26.46 → 26.00 → 25.48	
Observed 204.700 °C, Δ = 0.004 °C					Objective					25.0 ± 0.7 μm · writes 4 of 6	
Writes 1 of 3 budget, 0 rejected					Verdict					ATTAINED · 0.48 μm from target, 0.22 μm in band	
Verdict					every write opened a record judged keen						
VERIFICATION Governance, optimization and repeatability										separate archives, not pooled	
Four-arm governance ablation				Closed-loop controller				Evidence scope			
arm	rejected	breach	false blk								
Full	18/18	0	0	seed 42	70.167 → 86.845 (0.966)			Integrated	75 checks, 0 warn, 0 fail		
No interlock	12/18	6	0	seed 43	68.608 → 87.236 (0.970)			Governance	24/24 probes rejected, 0 false blocks		
No readback	18/18	0	0	seed 44	65.740 → 87.080 (0.969)			Portability	9/9 rejected, 0 code changes		
Both disabled	12/18	6	0	mean J/J*	0.968 (J* 89.894)			Pooled write p50	30.259 ms over 4 stacks		